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Editors
Review of Economic Dynamics

Dear Editors,

We are submitting "Loads, Not Bequests: An Order-Independent Decomposition of the Annuity Puzzle" for consideration at the *Review of Economic Dynamics*. The paper builds directly on a lineage the journal knows well: Lockwood's (2012, RED) bequest-motive analysis supplies our closest ancestor model and half our calibration, and our contribution is the tool his literature has lacked—an order-independent attribution across all the proposed channels at once.

Yaari (1965) proved that rational consumers should fully annuitize under actuarially fair pricing, yet voluntary ownership among US retirees is only a few percent. Six decades of research proposed competing "dominant" explanations—bequests, correlated health and mortality risk, pricing loads—but the sequential decompositions used to adjudicate them are order-dependent, so the field's answer has depended on which channel is switched on first. Pashchenko (2013, *JPubE*) came closest to a unified accounting but still predicted participation near 20%.

The paper nests nine rational and preference channels in one calibrated lifecycle model and attributes the annuity-demand gap with exact Shapley values over all $2^9 = 512$ channel subsets, so the attribution is order-independent by construction. The ranking is clear at the baseline and stable across the tested calibrations. Pricing loads are the dominant suppressor of demand (31.4 pp), with survival pessimism (10.9 pp) and the combined medical-expenditure and health-mortality channel (9.4 pp) as the co-leading second tier. Bequest motives, the single most-cited explanation in the literature, rank mid-pack (5.8 pp). Within the model, order-dependence alone can move the bequest attribution from 0 pp to 15.8 pp depending on where the channel enters the sequence, so the prior literature's disagreement over the dominant channel is at least partly an artifact of the sequential tool; the Shapley attribution removes that artifact rather than adding another entry to the argument.

A second contribution is methodological. The predicted ownership *level* is knife-edged in risk aversion—it moves from 0.0% to 21.9% as γ ranges over $[2.0, 3.0]$—because the extensive margin sits on a fixed-cost threshold, interacting with the pricing load, that the population crosses discontinuously. The same fragility plausibly explains why comparable models have predicted ownership anywhere from a few percent to full annuitization. The channel ranking, by contrast, is stable across that risk-aversion range, across wealth bands, and at alternative baseline money's worth ratios, with the boundary cases disclosed in the text. In calibrated annuity models, the ranking and the policy comparative statics, not the predicted level, are the objects on which to stake claims.

An application demonstrates the framework's policy reach. The same pricing load is the binding margin across the wealth distribution: at the baseline money's-worth ratio, immediate annuitization is value-destroying for all but the wealthiest, so predicted ownership rises with wealth and concentrates in the top wealth band (33.5%, against 0.0%in the bottom). Observed lifetime-annuity ownership also rises with wealth, but it is interior in the middle of the distribution, where the single-product model predicts none—a gap we attribute to better-priced employer and group annuities the model omits. The policy counterfactuals inherit this structure. A 22% Social Security benefit cut, the projected consequence of trust-fund exhaustion in late 2032, raises private annuity demand mainly in the top band (from 33.5% to 45.4%), while barely moving the lower-wealth households the cut most exposes. Retail longevity insurance does not backfill a public benefit cut where it bites hardest—and by-band counterfactuals show that even group or public-option pricing draws in mainly the top band; only pricing near actuarially fair levels reaches the middle of the distribution. Whether employer- or group-sourced annuities, which the observed interior ownership points to, could reach the exposed households is stated as the open question the single-product model leaves. The model's cross-sectional predictions are directionally consistent with the HRS data (5 of 7 channel-implied signs reproduced, though individually imprecise), and the structural prediction (7.9%) sits modestly above the two HRS ownership measures reported in parallel (3.64% on the lifetime-contract indicator, 6.06% on the any-annuity income proxy), neither of which enters the calibration.

Two exploratory behavioral channels, source-dependent utility and a narrow-framing at-purchase penalty, are reported separately. Calibrated to literature magnitudes rather than US moments, they move the prediction in opposite directions without cancelling—the penalty saturates the participation margin—and their inclusion reorders the structural attributions, which is exactly why the disciplined headline is the nine-channel structural game. We are explicit throughout that this extension is illustrative rather than identified.

The paper sits squarely in the journal's core genre—calibrated quantitative lifecycle analysis—and engages directly with Lockwood (2012, *RED*), Pashchenko (2013), and Peijnenburg, Nijman, and Werker (2016). Replication code in Julia and all calibration data are publicly available. The manuscript has not been submitted elsewhere and is not under consideration at another journal.

Sincerely,

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